

Syllabus: Higher-Order Metaphysics

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1 Official course description

Recent years have seen renewed interest in the use of higher-order languages as a vehicle for formulating metaphysical claims and arguments. After a brief introduction to the required formalism (which will be accessible to anyone familiar with first-order logic), this course will survey some recent work in the field. We will consider some object-level questions formulated in higher-order terms, including questions stated using the basic logical constants (e.g. whether $\forall p(p = \neg\neg p)$, where p is a variable of the same type as a sentence and $=$ expresses the analogue of identity for this type) as well as questions requiring further nonlogical constants relating to topics like necessity, grounding, fundamentality, qualitateness, meaning, and propositional attitudes. We will also discuss meta-level debates about the intelligibility or perspicuity of the higher-order formalism, as well as considering how claims expressed in higher-order terms might relate to claims expressed in English using words like ‘property’ and ‘proposition’. The readings will include some chapters from a forthcoming textbook by Andrew Bacon (Bacon, *A Philosophical Introduction to Higher-Order Logics*) and from a forthcoming collection edited by Peter Fritz and Nick Jones (Fritz and Jones, *Higher-Order Metaphysics*).

2 Assessment

If you are taking the course for credit, then—by default—you are expected to do some problem sets; turn in a **final paper** by the end of the winter break engaging with some of the themes of the seminar; and give a **presentation** on your paper idea during the last meeting of the seminar.

The goal of the problem sets is to get you comfortable using the formalism of higher order logic (in a few different versions), and understanding the statements and relevance of various results that will come up in the readings. You are permitted to work on them in collaboration with others, and encouraged to do so if you can find a collaboration that aids learning rather than impeding it. So long as you have made a reasonable effort with the problem sets, low scores on them will not lower your final grade. However, if you have done well on the problem sets, I will expect less of the final paper (in terms of length and ambition) than I would of a final paper in a regular graduate seminar with no other forms of assessment. For example, I’ll be more open to a final paper that just critiques one other paper.

The point of the final presentation is to give you an incentive to come up with a topic for the paper early enough to give a comfortable window of time to work on it, and to give some opportunity for feedback on the proposed topic. In addition, I strongly encourage you to discuss your proposed topic with me well in advance of the presentation.

Note that even though many of the meetings of the seminar will focus on relatively formal/technical material (on the grounds that these are the topics for which explanations by me are likely to be the most helpful), the topic of the seminar also provides plenty of scope for non-technical papers. For example, most of the papers in the forthcoming Fritz and Jones volume contain very little formalism; browsing the contents of this volume may be a helpful exercise if you are looking for inspiration in coming up with a topic.

3 Week-by-week topics and readings

This is just a rough indication and very likely to change. Many of the readings are drawn from the forthcoming textbook Bacon, *A Philosophical Introduction to Higher-Order Logics* (PIHOL). A draft of this will be available on Google Drive, where members of the seminar (including me) will be able to add comments and reply to comments by others. The draft we are using has some errors, which I'll add comments on if I notice them; if you spot something you think might be an error, please add a comment on it. If this goes well I will make other readings available for comment on Google Drive in the same way.

Week 1: Sep 6 The syntax of higher order logic: types and terms

Primary: PIHOL, ch. 0–3

Week 2: Sep 13 Four key systems of higher-order logic

Primary: PIHOL, ch. 4, 5, 6.4

Something on natural deduction version

Week 3: Sep 20 Prior's Theorem; the Russell-Myhill Theorem; arguments for beta-conversion.

Primary: Dorr, 'To Be F Is To Be G', §5–6

Supplementary: Bacon, Hawthorne and Uzquiano, 'Higher-Order Free Logic and the Prior-Kaplan Paradox'; Uzquiano, 'Ramified Structure'

Background: Russell, *The Principles of Mathematics*, Appendix B.; Myhill, 'Problems Arising in the Formalization of Intensional Logic'; Prior, 'On a Family of Paradoxes'.

Week 4: Sep 27 The Curry-Howard correspondence; product and sum types; reduction of product and sum types.

Primary: Geuvers, 'The Calculus of Constructions and Higher-Order Logic'; Dorr, 'Higher-Order Quantification and the Elimination of Abstract Objects', Appendix.

Week 5: Oct 4 Debating the intelligibility of higher-order languages

Primary: Sider, 'Higher-Order Metametaphysics', Hofweber, 'The Case Against Higher-Order Metaphysics', Bacon, 'A Case For Higher-Order Metaphysics', Goodman, 'Higher-order Logic as Metaphysics'.

Background: Prior, *Objects of Thought*, Rayo and Yablo, 'Nominalism through De/Nominalisation', Williamson, 'Everything'.

Week 6: Oct 11 How do formal higher-order languages relate to natural-language 'property'-talk?

Primary: Dorr, 'Higher-Order Quantification and the Elimination of Abstract Objects'.

Supplementary: Jones, 'Nominalist Realism', Button and Trueman, 'A Fictionalist Theory of Universals', Hofweber, 'The Case Against Higher-Order Metaphysics'.

Week 7: Oct 18 Model theory for higher-order logic.

Primary: PIHOL: part IV (excerpts). Or maybe I'll decide instead to use excerpts from Bacon and Dorr, 'Classicism'.

Week 8: Oct 25 ???

Week 9: Nov 1 Visit by Andrew Bacon (USC) (?)

Week 10: Nov 8 ???

Week 11: Nov 15 Visit by Jeremy Goodman (Johns Hopkins)

Week 12: Nov 29 ???

Week 13: Dec 6 ???

Week 14: Dec 13 Student presentations

For the four weeks with question marks, there is a range of possible topics we could discuss, the choice among which will depend in part on what people are interested in. Here are some possibilities:

- The treatment of propositional attitudes in higher order logic, and its relation to debates about the intelligibility/good standing of the formalism: Lederman, 'Higher-Order Metaphysics and Propositional Attitudes'.
- Higher-order logic and the debate about contingentism and necessitism Williamson, *Modal Logic as Metaphysics*, Fairchild, 'Symmetry and Hybrid Contingentism'.
- More powerful type systems, e.g. with dependent products and higher type universes. (I need to pick a good textbook treatment of this: there is a sketch in Crosilla, 'Constructive Type Theory, an Appetizer'.)
- Weaker (substructural) type systems and their metaphysical interest: PIHOL part III. (Or maybe Andrew will want to talk about this when he visits.)
- Theories of fundamentality in higher-order metaphysics: Dorr, 'To Be F Is To Be G', Bacon, 'Logical Combinatorialism'.
- "Maximalist" theories and their relatives, according to which—roughly—the only true identities are those provable in some given formal system (PIHOL 13, Bacon and Dorr, 'Classicism').
- The use of higher-order logic in defending minimalistic views about what individuals (type *e* things) there are. Can individuals even be eliminated altogether?

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